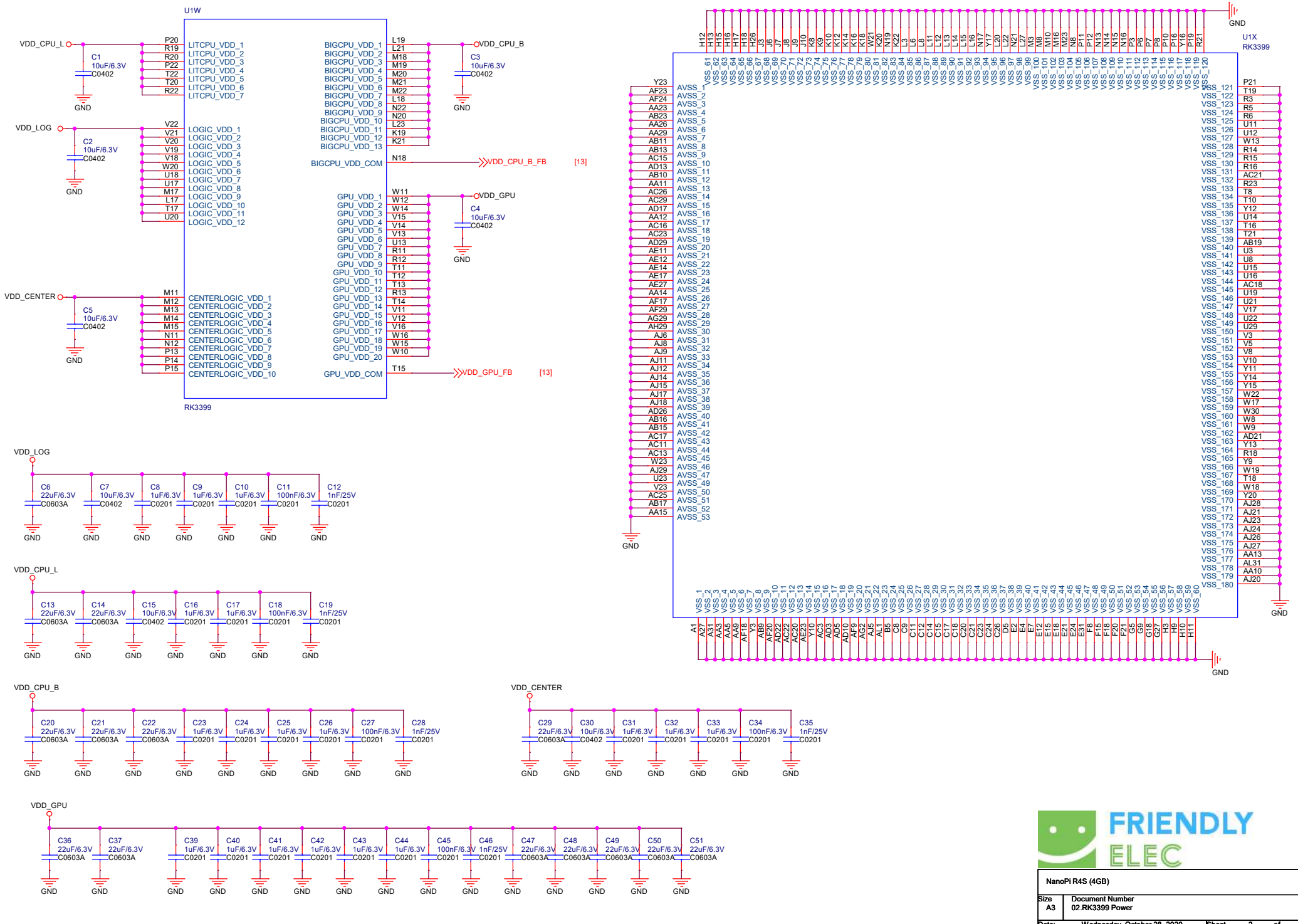
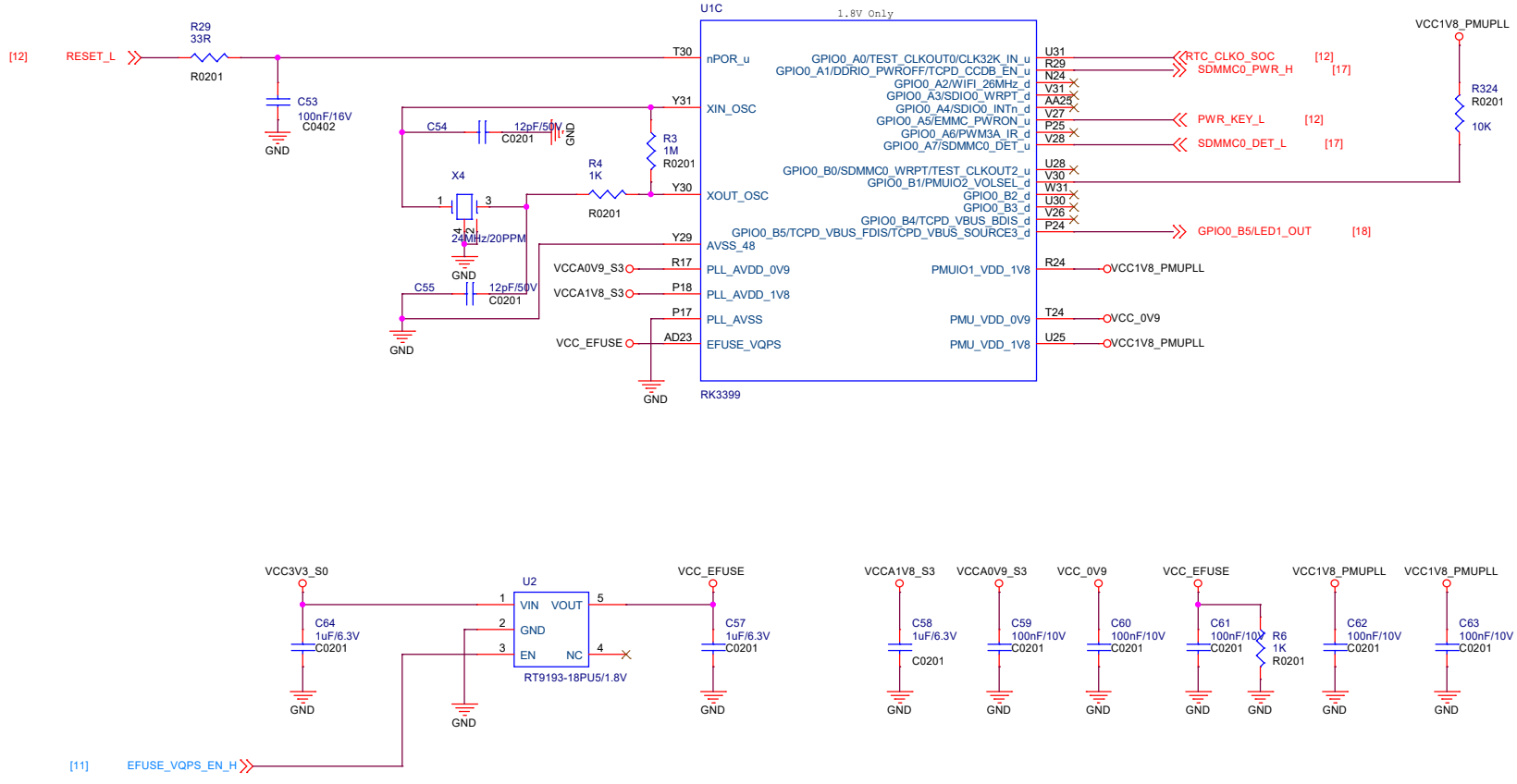
NanoPi R4S (4GB)

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NanoPi R4S (4GB)		
Size: A3	Document Number: 01.Title	Rev: 2008
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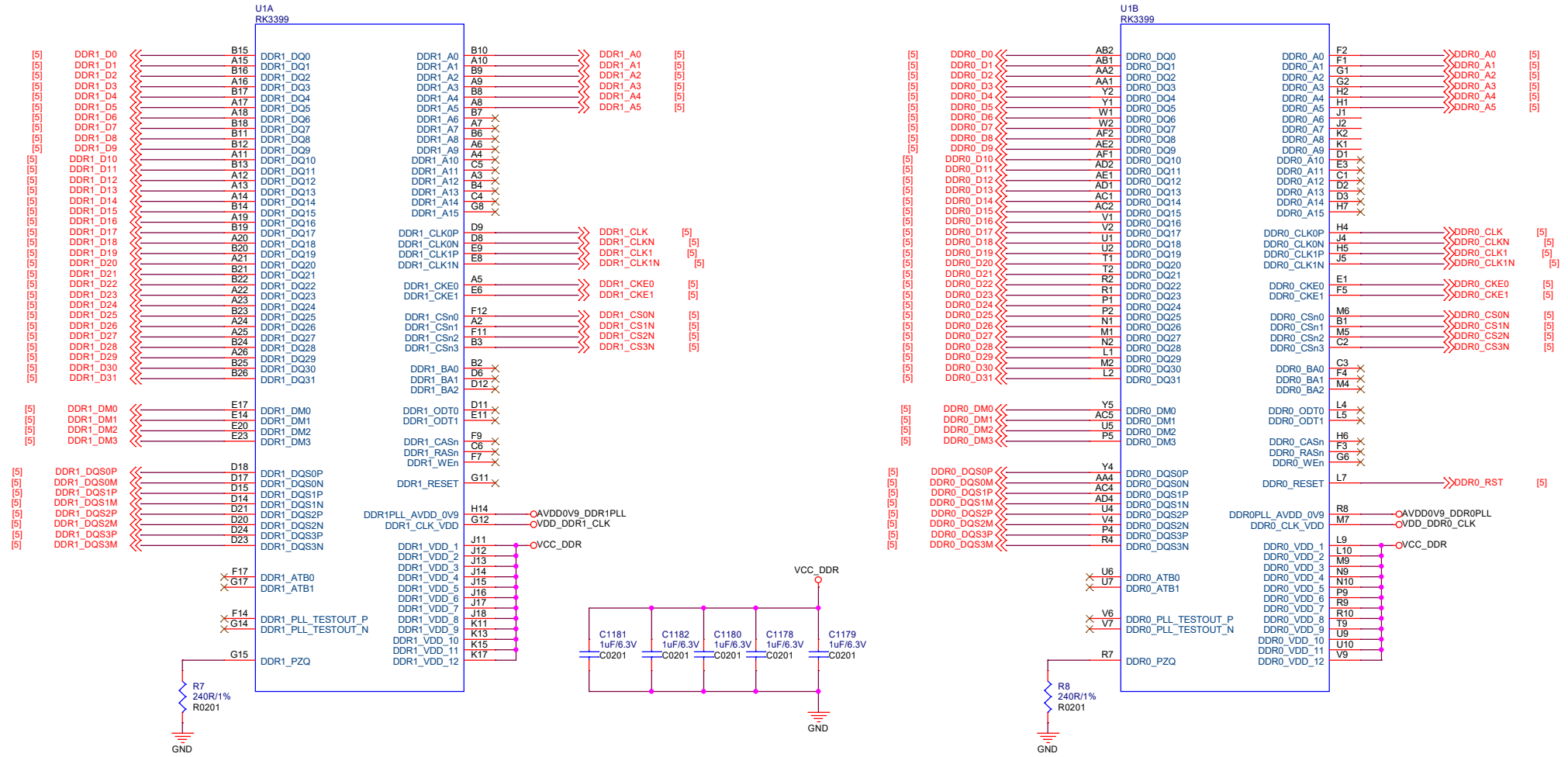


NanoPi R4S (4GB)	
Size: A3	Document Number: 02.RK3399 Power
Rev: 2008	
Date: Wednesday, October 28, 2020	Sheet 2 of 20

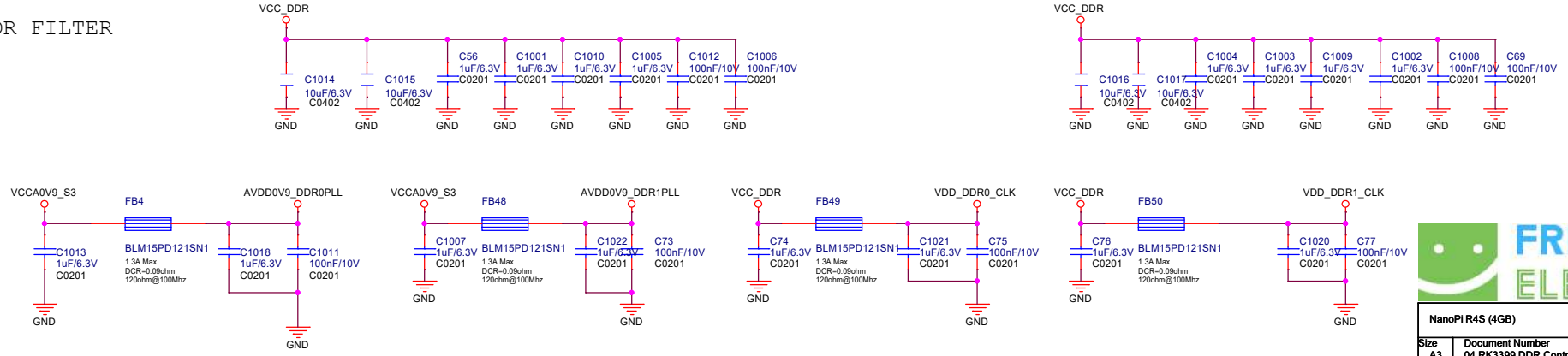


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Size A3	Document Number 03.RK3399 PMU Controler	Rev 2008
Date: Wednesday, October 28, 2020	Sheet 3 of 20	

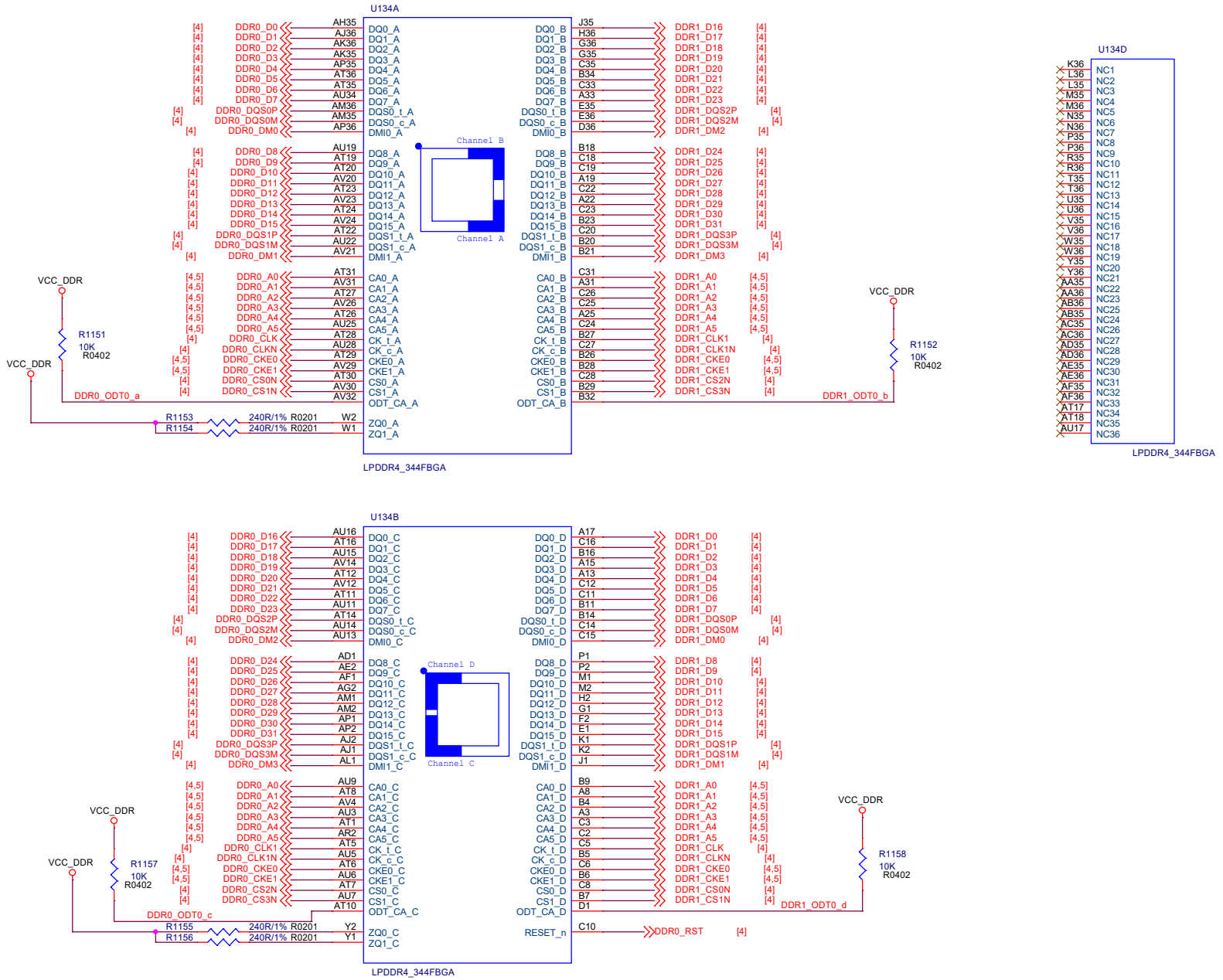


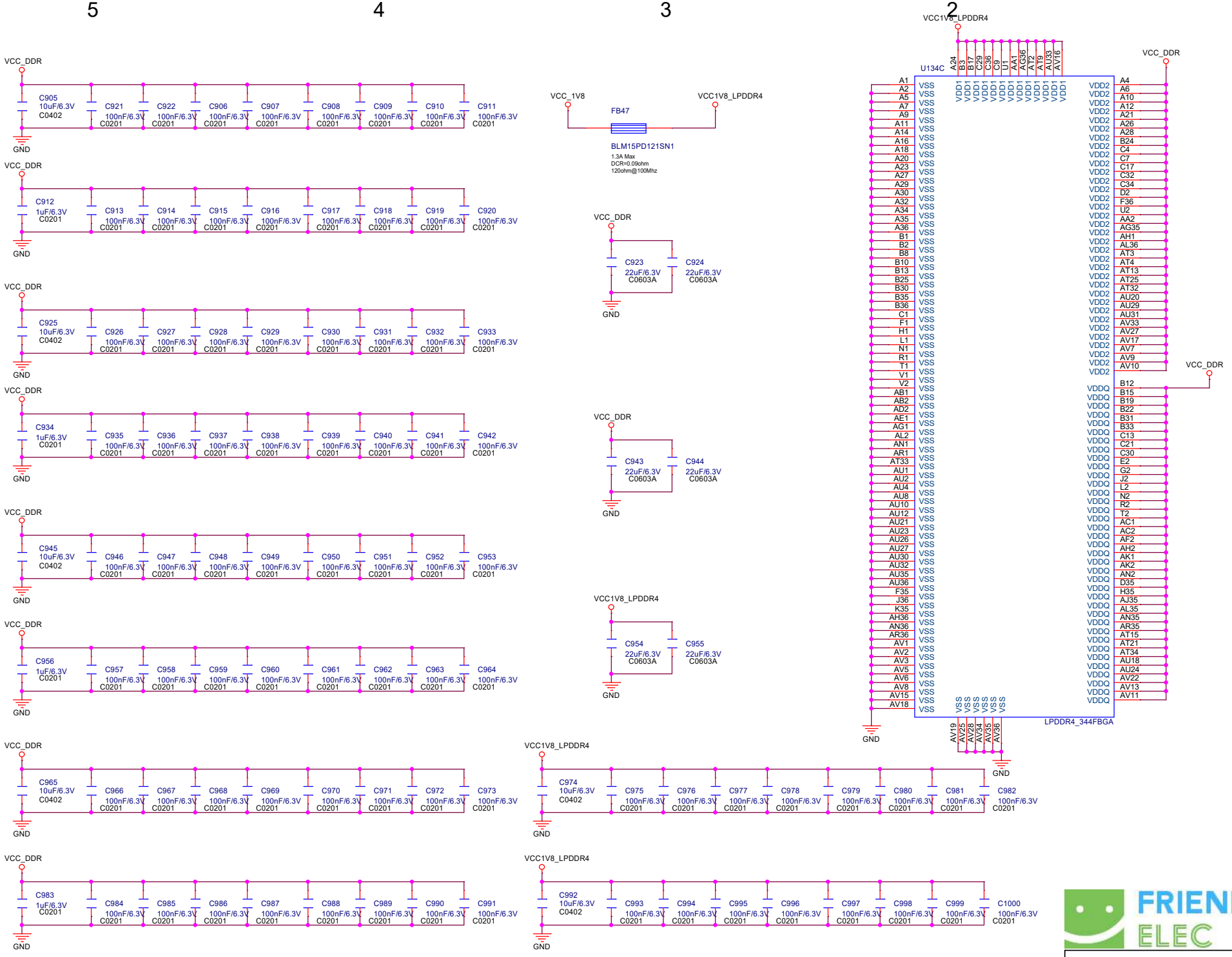
DDR FILTER



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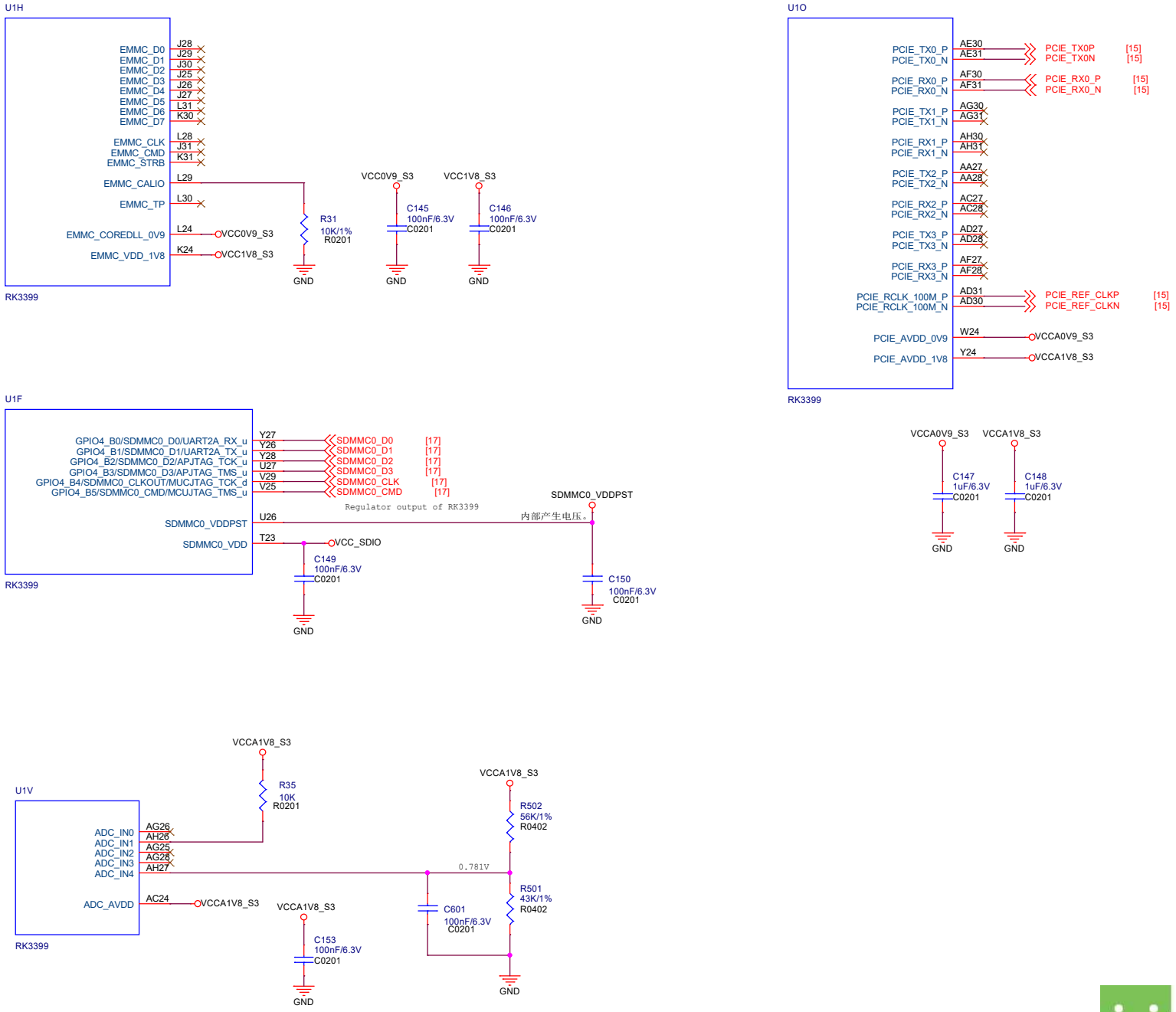
NanoPi R4S (4GB)		
Size: A3	Document Number: 04.RK3399 DDR Controler	Rev: 2008
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U134A
U134B
U134D
LPDDR4_344FBGA
Channel A
Channel B
Channel C
Channel D
VCC_DDR
R1151
R1152
R1153
R1154
R1155
R1156
R1157
R1158
10K
R0402
240R/1% R0201
DDR0_ODT0_a
DDR1_ODT0_b
DDR0_ODT0_c
DDR1_ODT0_d
DDR0_RST
NanoPi R4S (4GB)
Size A3
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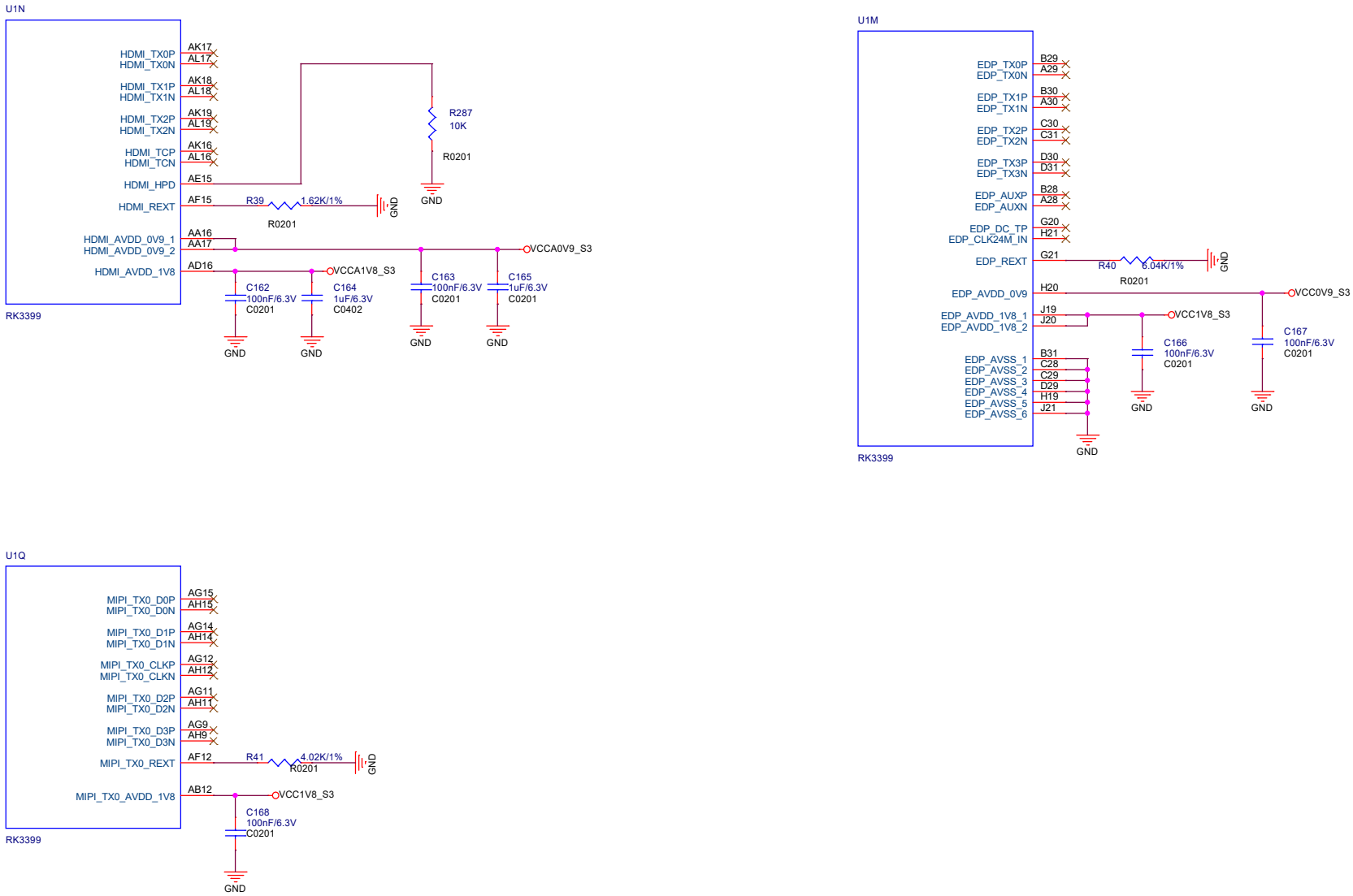
FRIENDLY ELEC

NanoPi R4S (4GB)		
Size A3	Document Number 06.LPDDR4_B	Rev 2008
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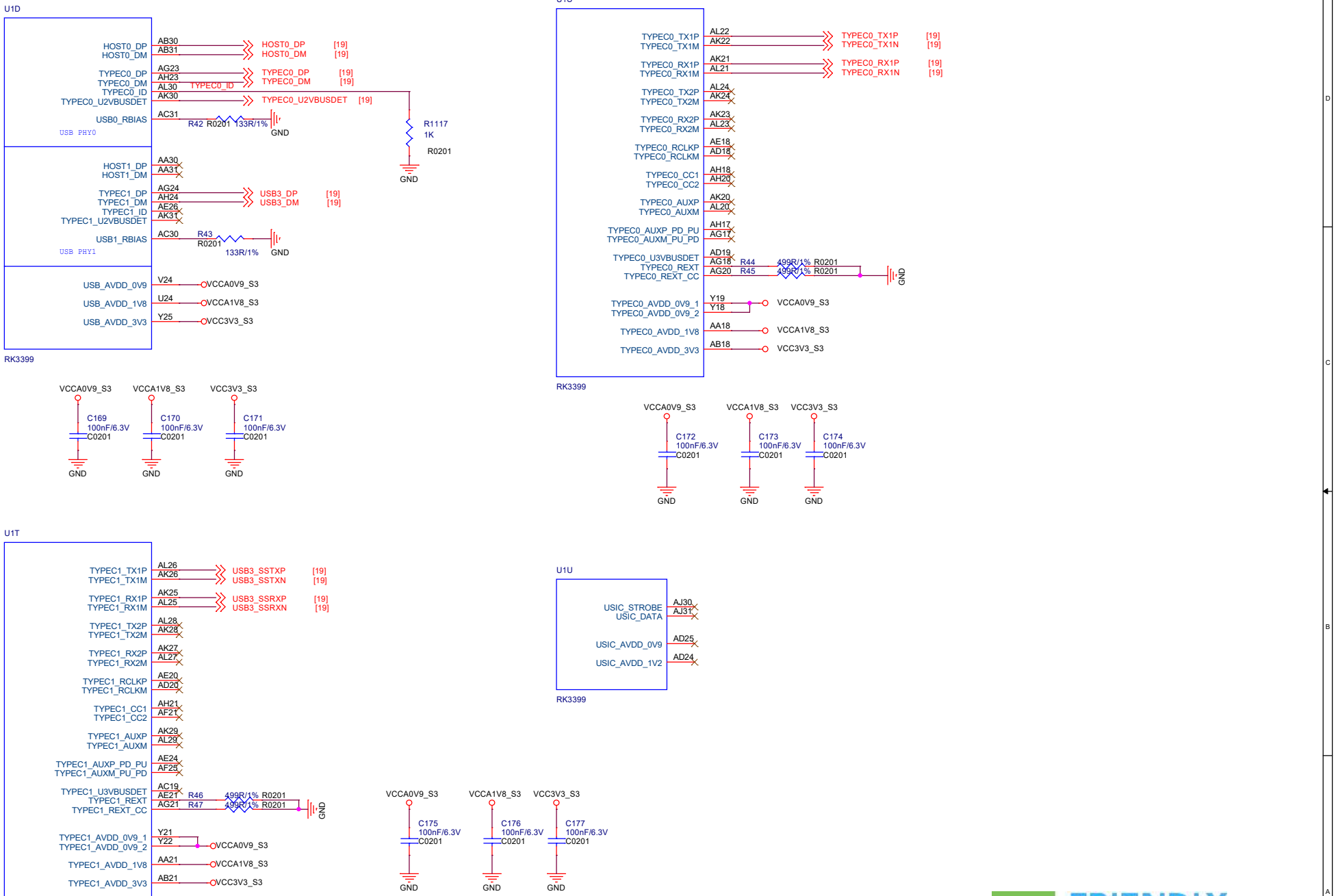
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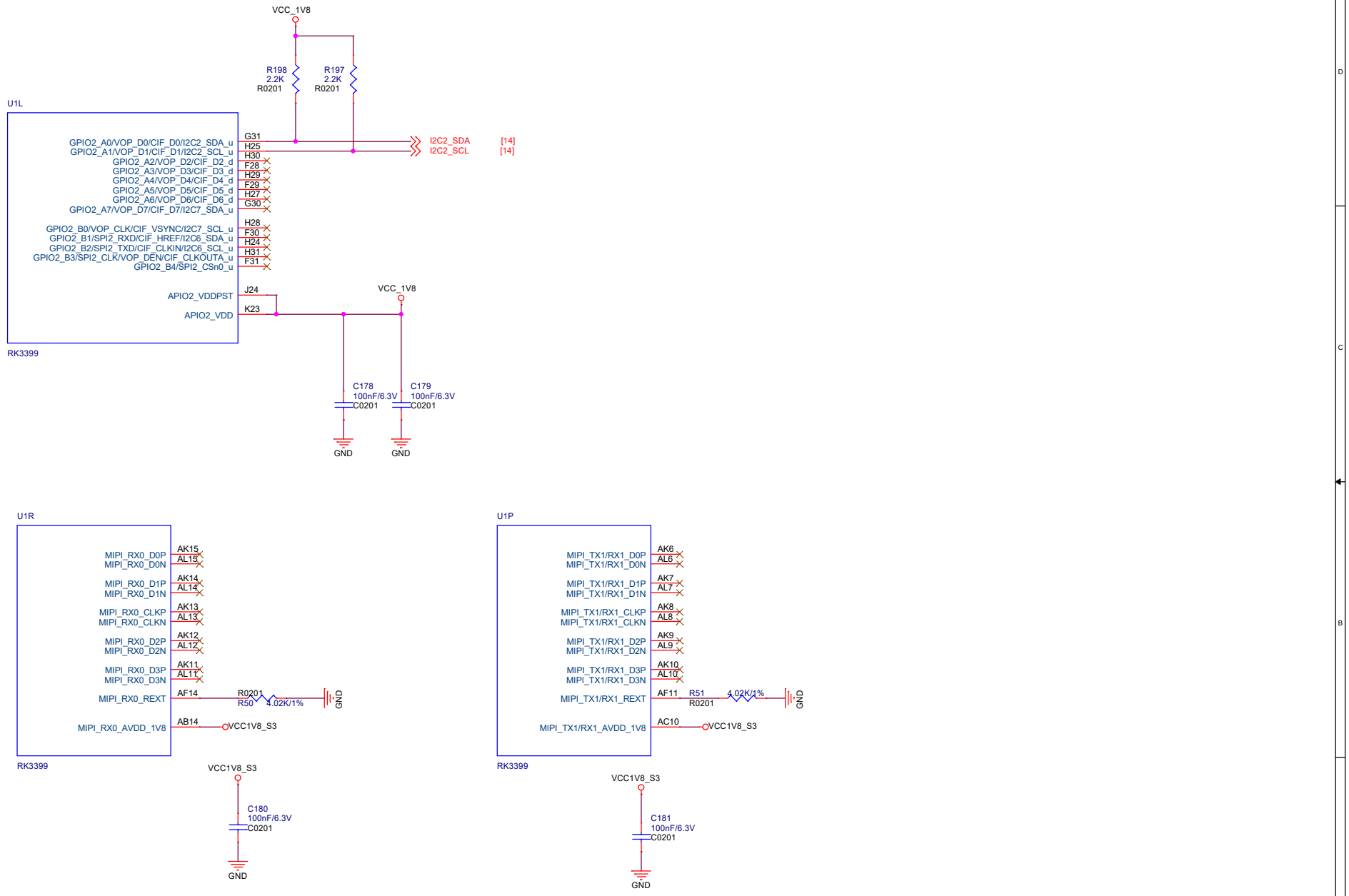
NanoPi R4S (4GB)		
Size A3	Document Number 06.RK3399 MMC/ADC/PCIe	Rev 2008
Date: Wednesday, October 28, 2020	Sheet 7 of 20	



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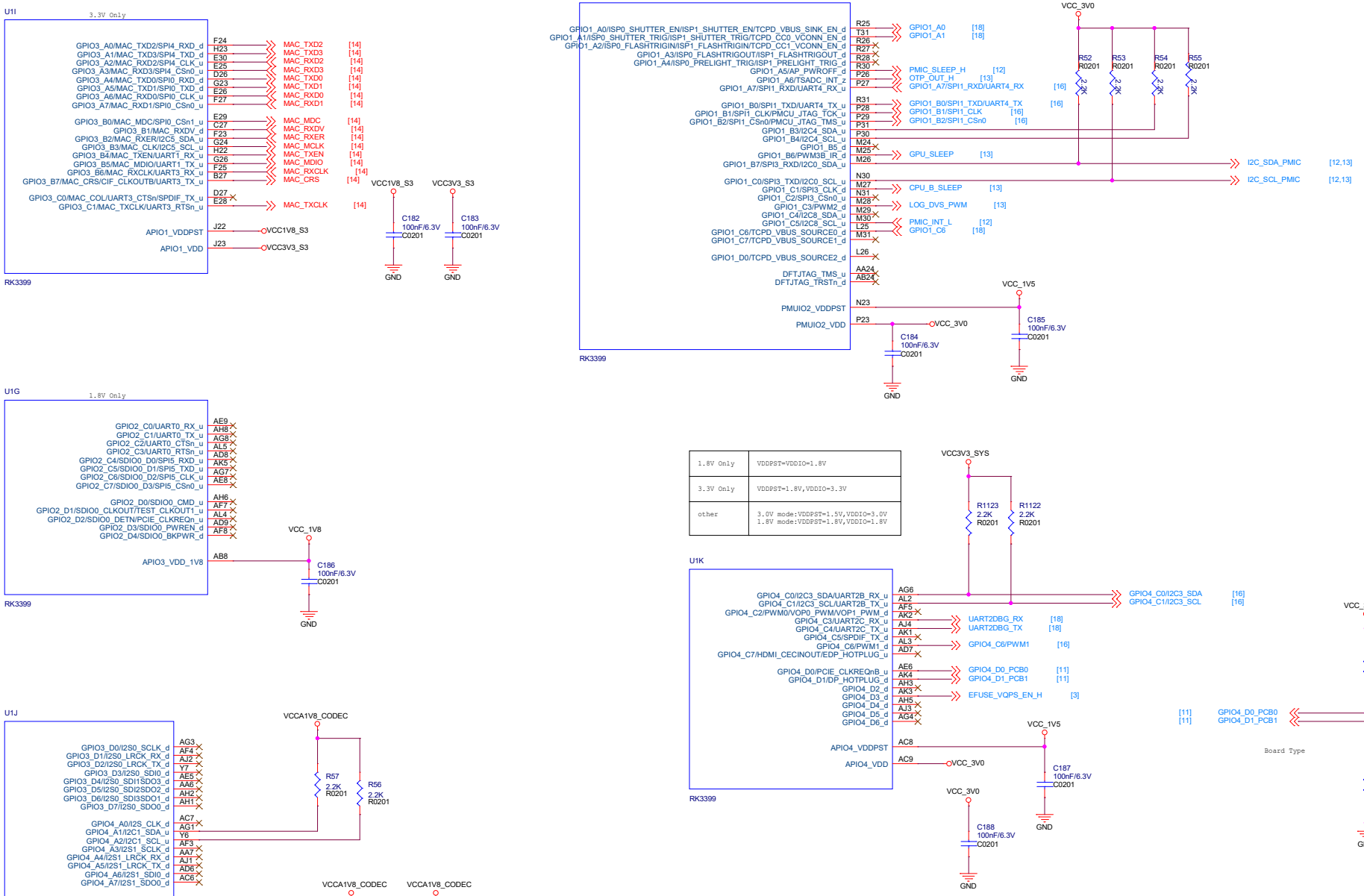
NanoPi R4S (4GB)		
Size A3	Document Number 07.RK3399 Display Interface	Rev 2008
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U1D
HOST0_DP
HOST0_DM
TYPEC0_DP
TYPEC0_DM
TYPEC0_ID
TYPEC0_U2VBUSDET
USB0_RBIAS
USB PHY0
AB30
AB31
AG23
AH23
AL30
AK30
AC31
HOST0_DP [19]
HOST0_DM [19]
TYPEC0_DP [19]
TYPEC0_DM [19]
TYPEC0_ID
TYPEC0_U2VBUSDET [19]
R42 R0201 133R/1%
GND
R1117
1K
R0201
GND
HOST1_DP
HOST1_DM
TYPEC1_DP
TYPEC1_DM
TYPEC1_ID
TYPEC1_U2VBUSDET
USB1_RBIAS
USB PHY1
AA30
AA31
AG24
AH24
AE26
AK31
AC30
USB3_DP [19]
USB3_DM [19]
R43
R0201
133R/1%
GND
USB_AVDD_0V9
USB_AVDD_1V8
USB_AVDD_3V3
V24
U24
Y25
VCCA0V9_S3
VCCA1V8_S3
VCC3V3_S3
RK3399
VCCA0V9_S3
VCCA1V8_S3
VCC3V3_S3
C169
100nF/6.3V
C0201
C170
100nF/6.3V
C0201
C171
100nF/6.3V
C0201
GND
GND
GND
U1S
TYPEC0_TX1P
TYPEC0_TX1M
TYPEC0_RX1P
TYPEC0_RX1M
TYPEC0_TX2P
TYPEC0_TX2M
TYPEC0_RX2P
TYPEC0_RX2M
TYPEC0_RCLKP
TYPEC0_RCLKM
TYPEC0_CC1
TYPEC0_CC2
TYPEC0_AUXP
TYPEC0_AUXM
TYPEC0_AUXP_PD_PU
TYPEC0_AUXM_PU_PD
TYPEC0_U3VBUSDET
TYPEC0_REXT
TYPEC0_REXT_CC
TYPEC0_AVDD_0V9_1
TYPEC0_AVDD_0V9_2
TYPEC0_AVDD_1V8
TYPEC0_AVDD_3V3
AL22
AK22
AK21
AL21
AL24
AK24
AK23
AL23
AE18
AD18
AH18
AH20
AK20
AL20
AH17
AG17
AD19
AG18
AG20
R44
R45
499R/1% R0201
499R/1% R0201
GND
Y19
Y18
AA18
AB18
TYPEC0_TX1P [19]
TYPEC0_TX1N [19]
TYPEC0_RX1P [19]
TYPEC0_RX1N [19]
VCCA0V9_S3
VCCA1V8_S3
VCC3V3_S3
RK3399
VCCA0V9_S3
VCCA1V8_S3
VCC3V3_S3
C172
100nF/6.3V
C0201
C173
100nF/6.3V
C0201
C174
100nF/6.3V
C0201
GND
GND
GND
U1T
TYPEC1_TX1P
TYPEC1_TX1M
TYPEC1_RX1P
TYPEC1_RX1M
TYPEC1_TX2P
TYPEC1_TX2M
TYPEC1_RX2P
TYPEC1_RX2M
TYPEC1_RCLKP
TYPEC1_RCLKM
TYPEC1_CC1
TYPEC1_CC2
TYPEC1_AUXP
TYPEC1_AUXM
TYPEC1_AUXP_PD_PU
TYPEC1_AUXM_PU_PD
TYPEC1_U3VBUSDET
TYPEC1_REXT
TYPEC1_REXT_CC
TYPEC1_AVDD_0V9_1
TYPEC1_AVDD_0V9_2
TYPEC1_AVDD_1V8
TYPEC1_AVDD_3V3
AL26
AK26
AK25
AL25
AL28
AK28
AK27
AL27
AE20
AD20
AH21
AF21
AK29
AL29
AE24
AF25
AC19
AE21
AG21
R46
R47
499R/1% R0201
499R/1% R0201
GND
Y21
Y22
AA21
AB21
USB3_SSTXP [19]
USB3_SSTXN [19]
USB3_SSRXP [19]
USB3_SSRXN [19]
VCCA0V9_S3
VCCA1V8_S3
VCC3V3_S3
RK3399
VCCA0V9_S3
VCCA1V8_S3
VCC3V3_S3
C175
100nF/6.3V
C0201
C176
100nF/6.3V
C0201
C177
100nF/6.3V
C0201
GND
GND
GND
U1U
USIC_STROBE
USIC_DATA
USIC_AVDD_0V9
USIC_AVDD_1V2
AJ30
AJ31
AD25
AD24
RK3399
NanoPi R4S (4GB)
Size A3
Document Number 08.RK3399 USB/USIC Controler
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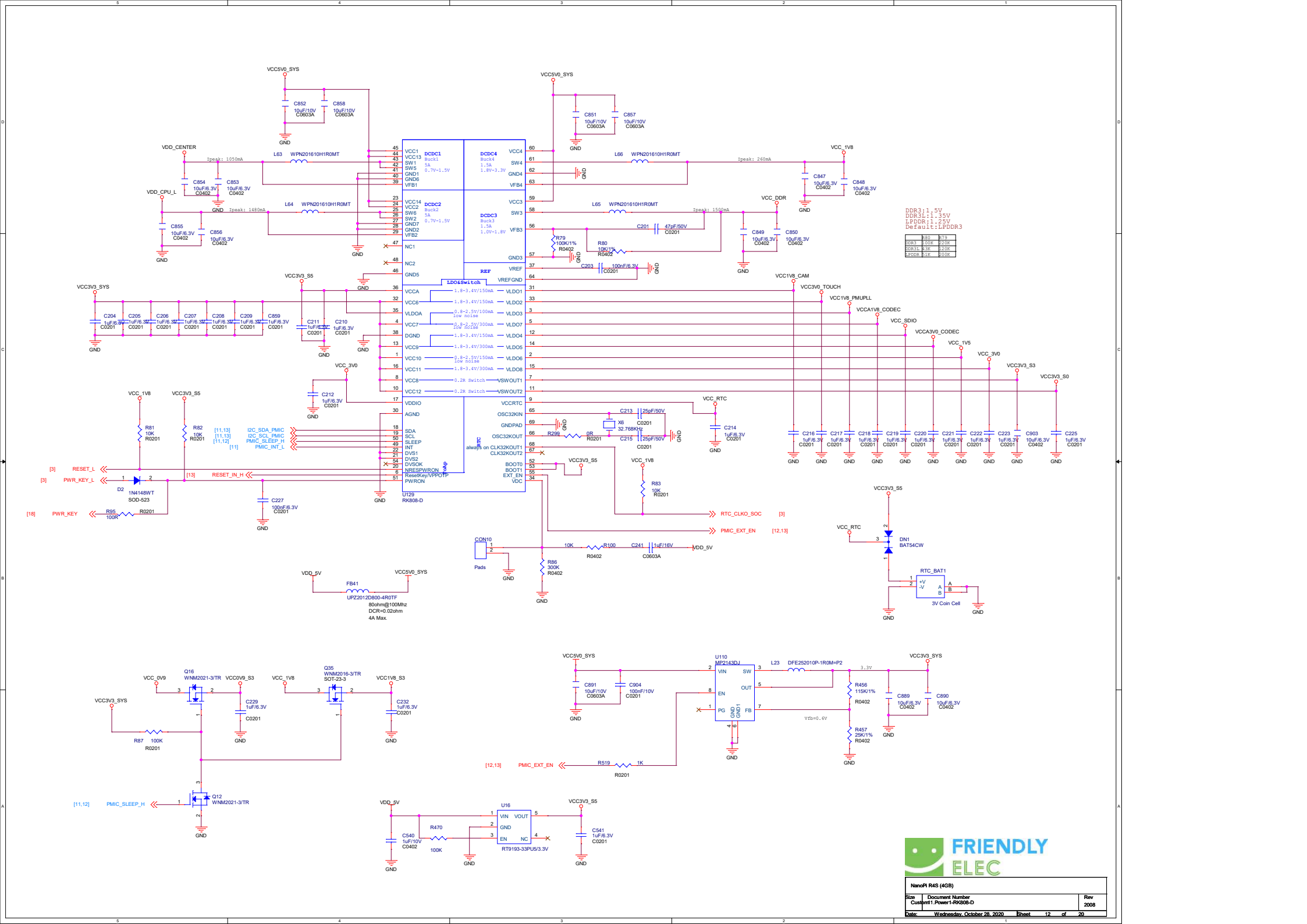
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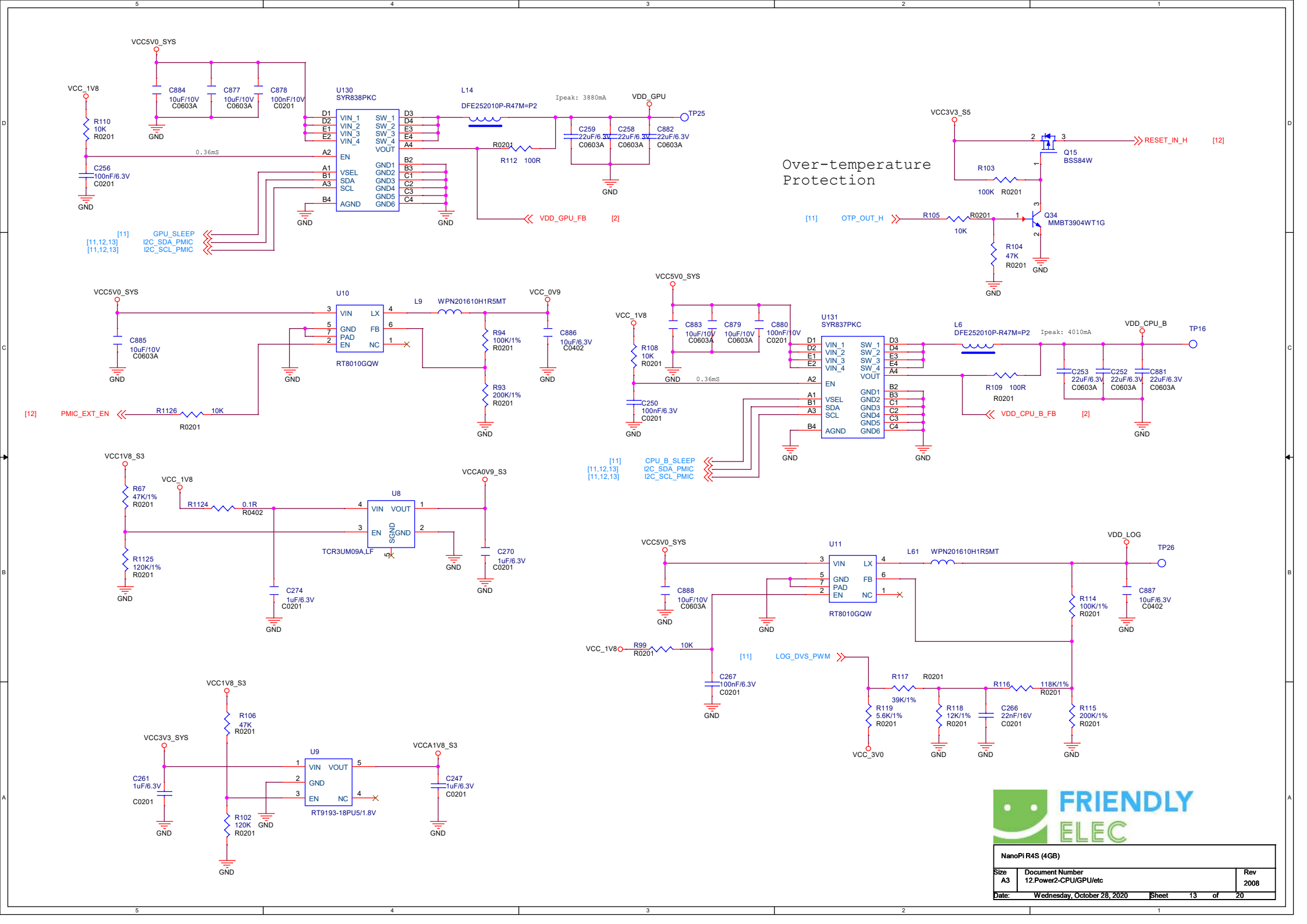
NanoPi R4S (4GB)		
Size A3	Document Number 09.RK3399 DVP/MIPI CSI	Rev 2008
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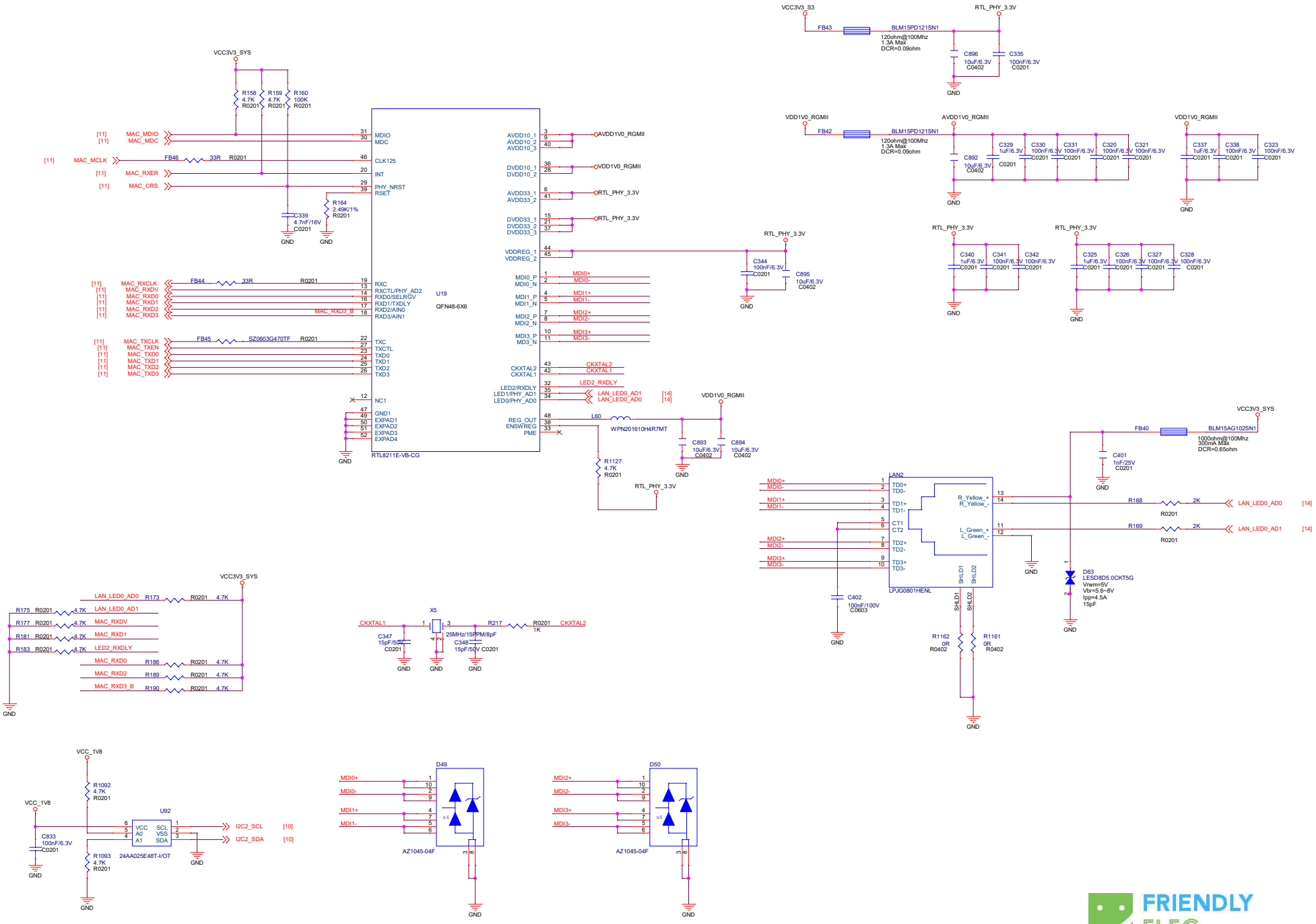
NanoPi R4S (4GB)		
Size: Custom	Document Number: 10.RK3399 RGMII/SDIO/SPI/etc	Rev: 2008
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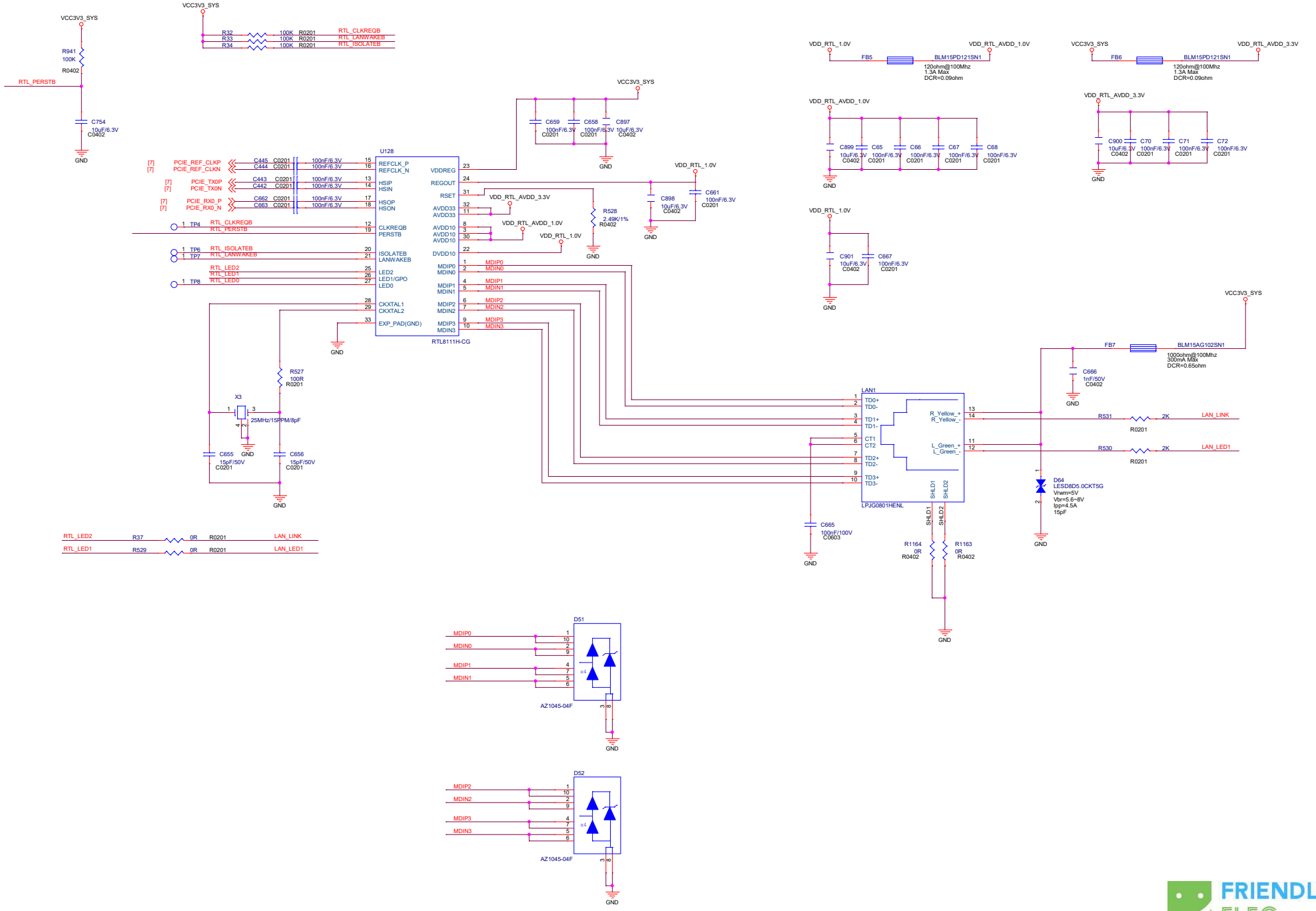

VCC5V0_SYS
C852 10uF/10V C0603A
C858 10uF/10V C0603A
GND
VDD_CENTER
Ipeak: 1050mA
L63 WPN201610H1R0MT
C854 10uF/6.3V C0402
C853 10uF/6.3V C0402
VDD_CPU_L
Ipeak: 1480mA
L64 WPN201610H1R0MT
C855 10uF/6.3V C0402
C856 10uF/6.3V C0402
VCC1 VCC13 SW1 SW5 GND1 GND6 VFB1
DCDC1 Buck1 5A 0.7V-1.5V
VCC14 VCC2 SW6 SW2 GND7 GND2 VFB2
DCDC2 Buck2 5A 0.7V-1.5V
NC1
NC2
GND5
VCCA VCC6 VLDOA VCC7 DGND VCC9 VCC10 VCC11 VCC8 VCC12
LDO&Switch
1.8-3.4V/150mA VLDO1
1.8-3.4V/150mA VLDO2
0.8-2.5V/100mA low noise VLDO3
0.8-2.5V/300mA low noise VLDO7
1.8-3.4V/150mA VLDO4
1.8-3.4V/300mA VLDO5
0.8-2.5V/150mA low noise VLDO6
1.8-3.4V/300mA VLDO8
0.2R Switch VSWOUT1
0.2R Switch VSWOUT2
VDDIO AGND
SDA SCL SLEEP INT DVS1 DVS2 DVSOK NRESPWRON ResetKey/VPPOTP PWRON
U129 RK808-D
DCDC4 Buck4 1.5A 1.8V-3.3V VCC4 SW4 GND4 VFB4
DCDC3 Buck3 1.5A 1.0V-1.8V VCC3 SW3 GND3 VFB3
REF VREF VREFGND
RTC always on CLK32KOUT1 CLK32KOUT2
OSC32KIN GNDPAD OSC32KOUT
BOOT0 BOOT1 EXT_EN VDC
VCC5V0_SYS
C851 10uF/10V C0603A
C857 10uF/10V C0603A
L66 WPN201610H1R0MT
Ipeak: 260mA
VCC_1V8
C847 10uF/6.3V C0402
C848 10uF/6.3V C0402
L65 WPN201610H1R0MT
Ipeak: 1500mA
VCC_DDR
C201 47pF/50V C0201
C849 10uF/6.3V C0402
C850 10uF/6.3V C0402
R79 100K/1% R0402
R80 10K/1% R0402
C203 100nF/6.3V C0201
DDR3:1.5V
DDR3L:1.35V
LPDDR:1.25V
Default:LPDDR3
R80 R79
DDR3 100K 220K
DDR3L 43K 120K
LPDDR 51K 200K
VCC3V3_SYS
C204 C205 C206 C207 C208 C209 C859 1uF/6.3V C0201
VCC3V3_S5
C211 1uF/6.3V C0201
C210 1uF/6.3V C0201
VCC_3V0
C212 1uF/6.3V C0201
VCC1V8_CAM
VCC3V0_TOUCH
VCC1V8_PMUPLL
VCCA1V8_CODEC
VCC_SDIO
VCCA3V0_CODEC
VCC_1V5
VCC_3V0
VCC3V3_S3
VCC3V3_S0
C216 C217 C218 C219 C220 C221 C222 C223 C903 10uF/6.3V C0402
C225 1uF/6.3V C0201
VCC_RTC
C213 25pF/50V C0201
X6 32.768KHz
C215 25pF/50V C0201
R299 0R R0201
C214 1uF/6.3V C0201
R81 10K R0201
R82 10K R0201
[11,13] I2C_SDA_PMIC
[11,13] I2C_SCL_PMIC
[11,12] PMIC_SLEEP_H
[11] PMIC_INT_L
[3] RESET_L
[3] PWR_KEY_L
[13] RESET_IN_H
D2 1N4148WT SOD-523
[18] PWR_KEY
R95 100R
R0201
C227 100nF/6.3V C0201
R83 10K R0201
RTC_CLKO_SOC [3]
PMIC_EXT_EN [12,13]
CON10 Pads
R100 10K R0402
C241 1uF/16V C0603A
VDD_5V
R86 300K R0402
FB41 UPZ2012D800-4R0TF
80ohm@100Mhz
DCR=0.02ohm
4A Max
VCC3V3_S5
DN1 BAT54CW
RTC_BAT1
3V Coin Cell
Q16 WNM2021-3/TR
VCC_0V9
VCC0V9_S3
C229 1uF/6.3V C0201
Q35 WNM2016-3/TR SOT-23-3
VCC1V8_S3
C232 1uF/6.3V C0201
R87 100K R0201
Q12 WNM2021-3/TR
[11,12] PMIC_SLEEP_H
U110 MP2143DJ
VIN SW OUT EN PG FB GND GND1
L23 DFE252010P-1R0M=P2
3.3V
R456 115K/1% R0402
R457 25K/1% R0402
Vfb=0.6V
C889 10uF/6.3V C0402
C890 10uF/6.3V C0402
C891 10uF/10V C0603A
C904 100nF/10V C0201
R519 1K R0201
U16 RT9193-33PU5/3.3V
VIN VOUT GND EN NC
R470 100K
C540 1uF/10V C0402
C541 1uF/6.3V C0201
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NanoPi R4S (4GB)
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Size A3	Document Number 12.Power2-CPU/GPU/etc	Rev 2008
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10/100/1000M Ethernet

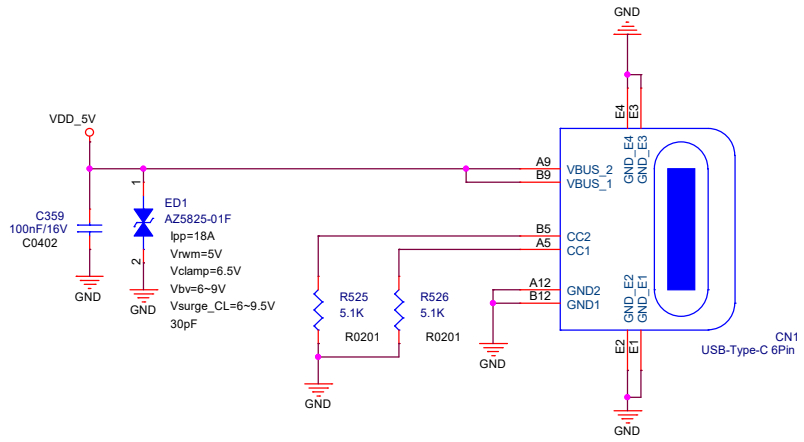


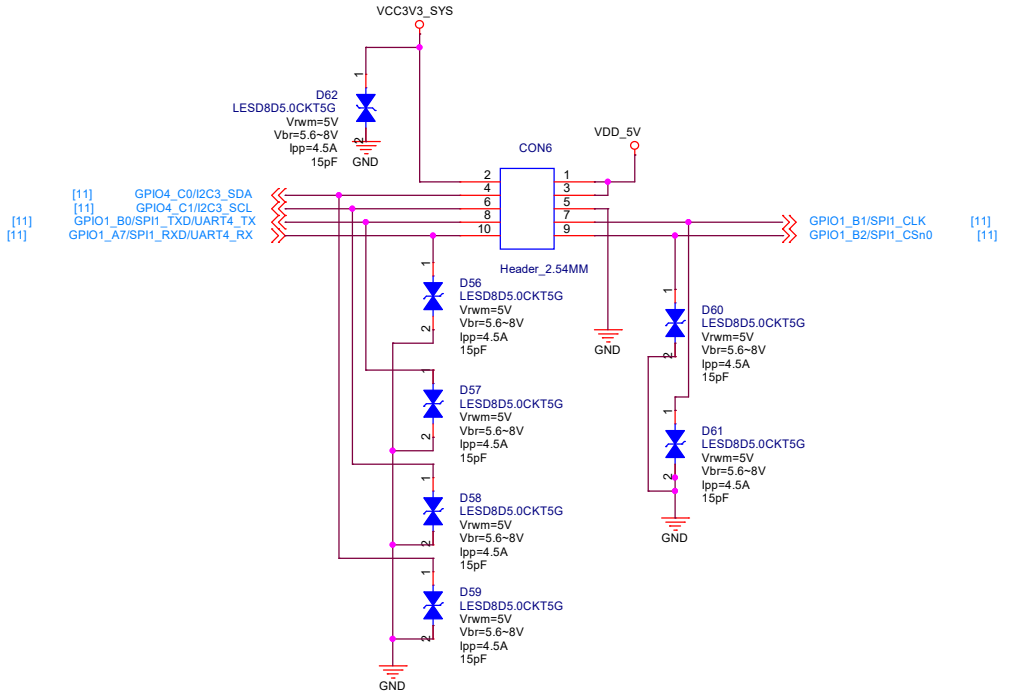


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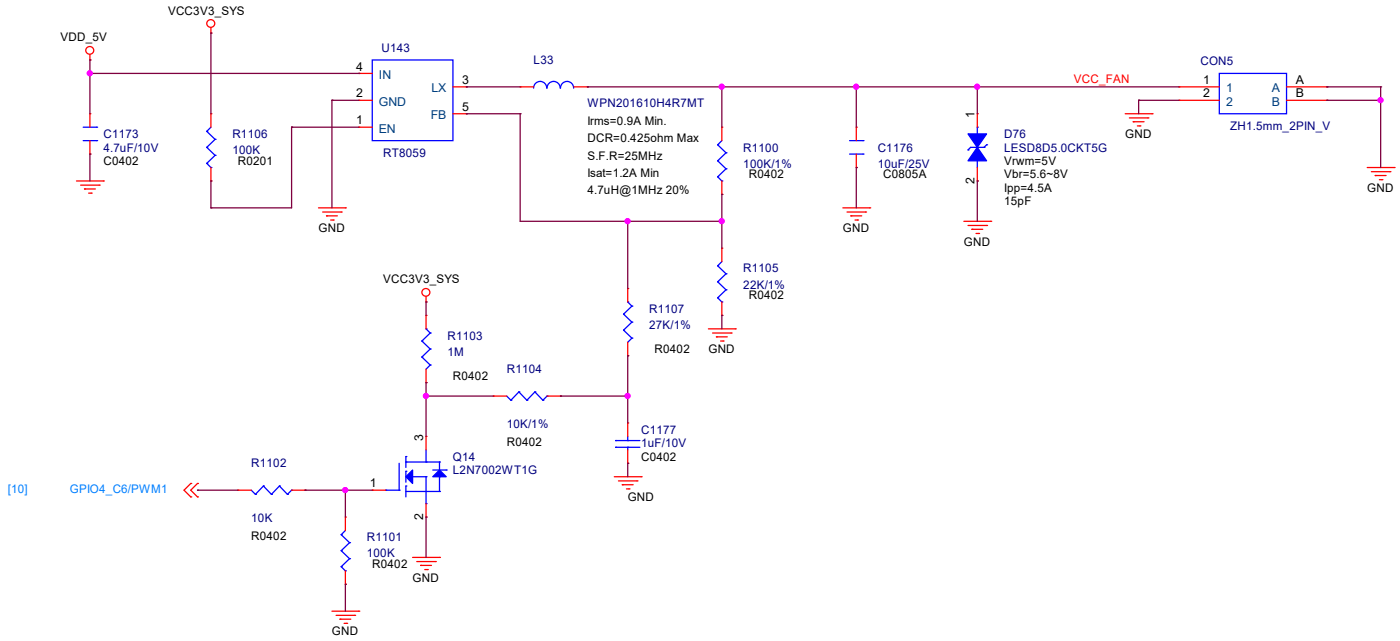
NanoPI R4S (4GB)		
Size A2	Document Number 15.PCIe LAN	Rev 2008
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USB Type-C



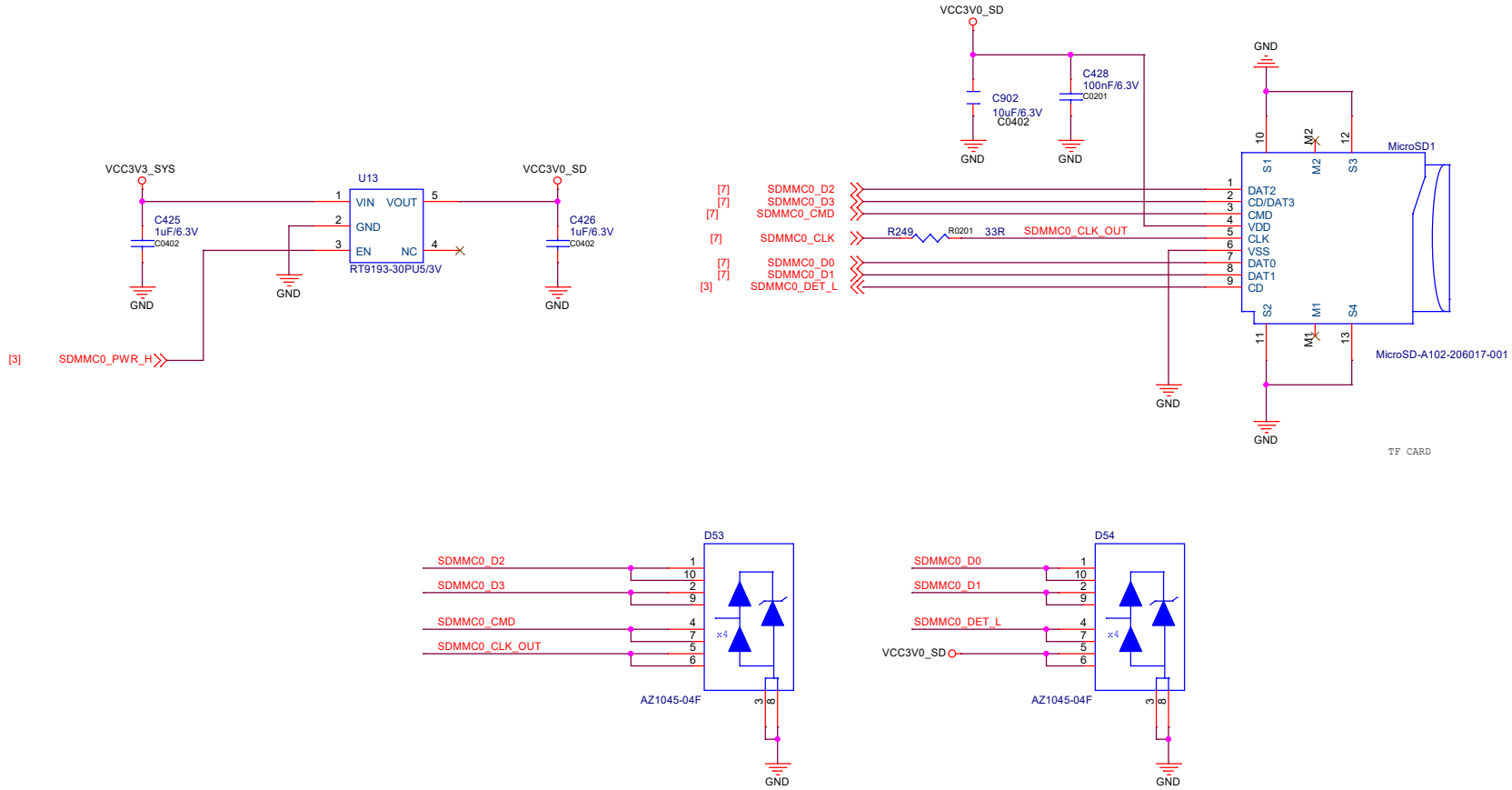


FAN

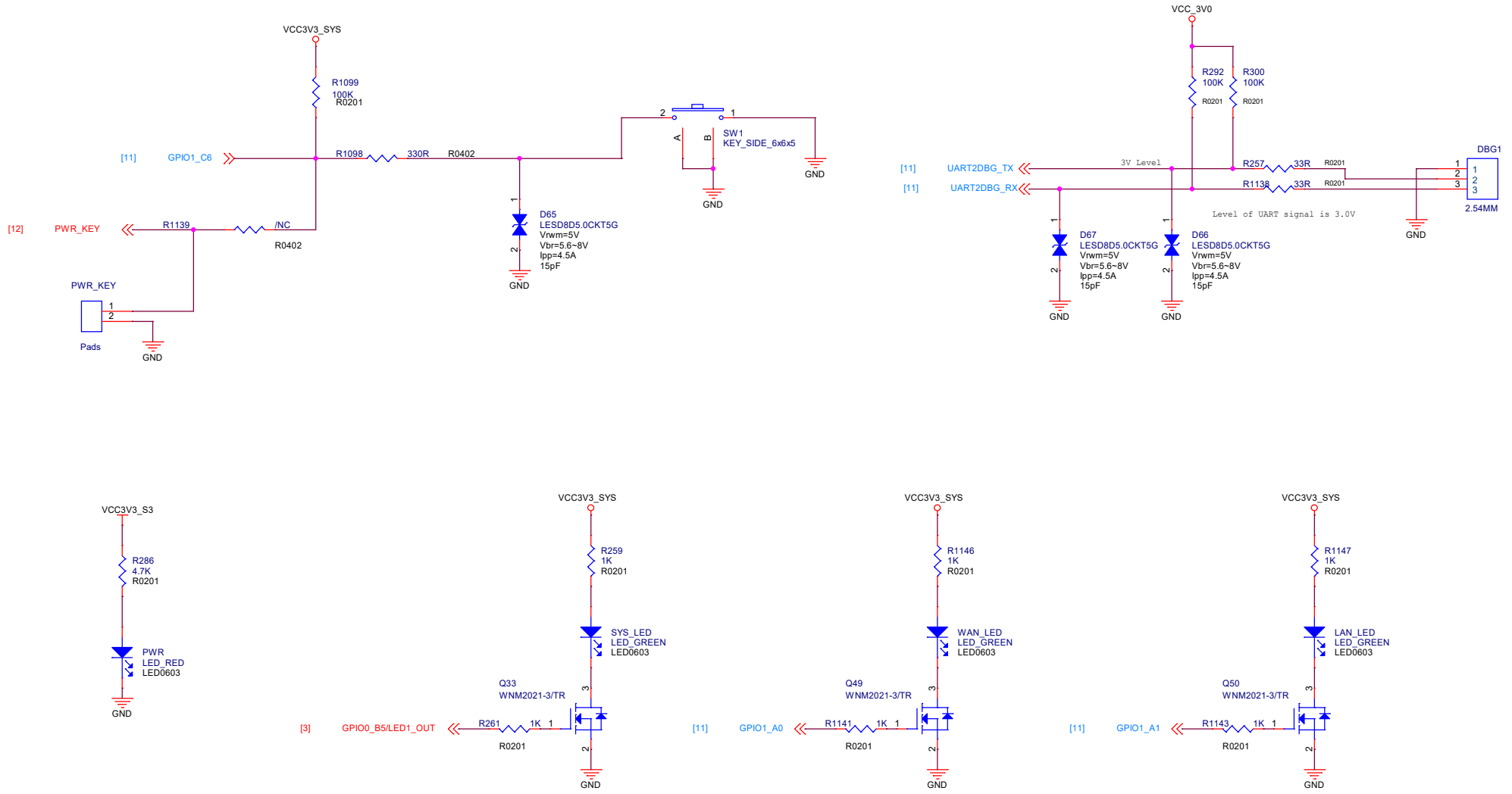


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NanoPi R4S (4GB)		
Size A3	Document Number 16.USB Type-C	Rev 2008
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NanoPi R4S (4GB)		
Size A3	Document Number 17.SD Card	Rev 2008
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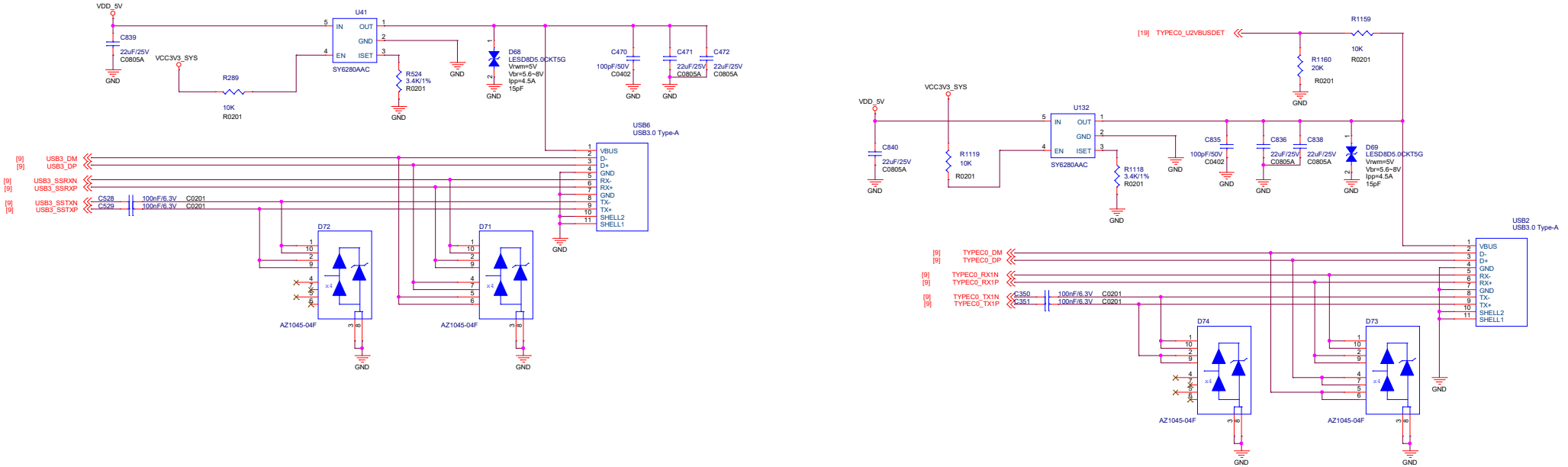


NanoPi R4S (4GB)		
Size A3	Document Number 18.Key/LED/Debug	Rev 2008
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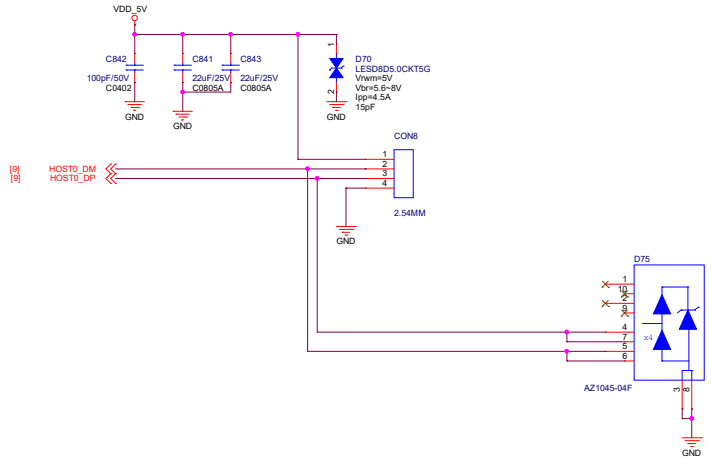
5 4 3 2 1

D C B A

USB3.0 Host



USB2.0 Host

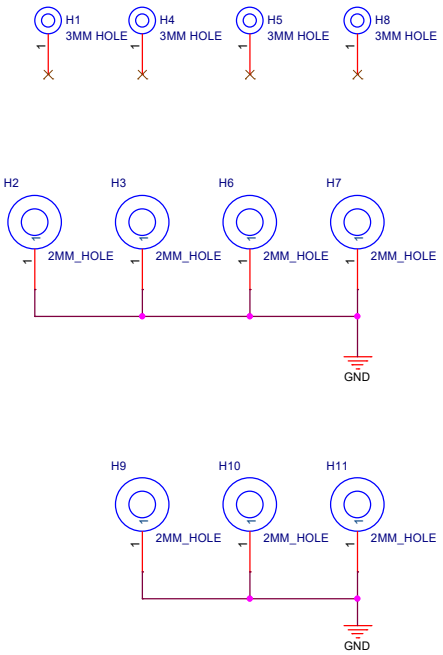


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